PRE-CALCULUS
NUMBER SYSTEMS

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Number systems

The **integers** are the set of all positive and negative whole numbers, and zero:

$$\ldots, -3, -2, -1, 0, 1, 2, 3, 4, \ldots$$

The set of integers is denoted by $\mathbb{Z}$. From the integers we can construct the **rational numbers**. These are all the ratios of integers. That is, any rational number r can be written as

$$r = \frac{m}{n} \text{ where } m, n \in \mathbb{Z}, n \neq 0.$$

The following are examples of rational numbers (elements of $\mathbb{Q}$):

$$\frac{1}{3} \qquad \frac{22}{7} \qquad \frac{-6}{7} \qquad 3 = \frac{24}{8} = \frac{3}{1}$$

$$1.72 = \frac{172}{100} = \frac{43}{25} \qquad 1 = \frac{1}{1}$$

Number systems

Some numbers cannot be written as $\frac{m}{n}$ for $m, n \in \mathbb{Z}$. These are the *irrational* numbers.

$$\sqrt{2} \qquad \sqrt[3]{9} \qquad \pi \qquad e \qquad \log_{10} 2$$

Combining the rational and irrational numbers gives us the set of *real numbers*, denoted $\mathbb{R}$. Every $x \in \mathbb{R}$ has a decimal expansion. For rationals, the decimal will start to repeat at some point. For example:

$$\frac{1}{3} = 0.33333\ldots = 0.\overline{3} \qquad \frac{1}{7} = 0.\overline{142857}$$

The real numbers

The real numbers are **totally ordered**. We can compare any two real numbers and say whether the first one is bigger than the second one, whether the second is bigger than the first, or whether they are equal.

The following are examples of true inequalities:

$$7 < 7.4 < 7.5 \qquad -\pi < -3 \qquad \sqrt{2} < 2$$

$$\sqrt{2} \leqslant 2 \qquad 2 \leqslant 2 \qquad -10 < \sqrt{100}$$

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SET NOTATION

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Set notation

A **set** is a collection of objects. If S is a set, we write $a \in S$ to say that a is an element of S. We can also write $a \notin S$ to mean that a is *not* an element of S.

Example: $3 \in \mathbb{Z}$ but $\pi \notin \mathbb{Z}$.

Example of set-builder notation:

$$\begin{aligned} A &= \{1, 2, 3, 4, 5, 6\} \\ &= \{x \in \mathbb{Z} \mid 0 < x < 7\} \end{aligned}$$

Another example of set notation

$$
\begin{aligned}
\{-2,-1,0,1,2,3\} &= \{x \in \mathbb{Z} \mid -2 \leqslant x \leqslant 3\} \\
&= \{x \in \mathbb{Z} \mid -3 < x \leqslant 3\} \\
&= \{x \in \mathbb{Z} \mid -3 < x < 4\} \\
&= \{x \in \mathbb{Z} \mid -2 \leqslant x < 4\}
\end{aligned}
$$

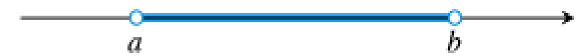
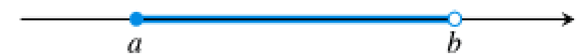
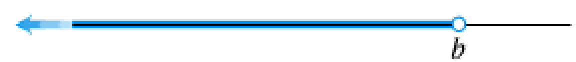
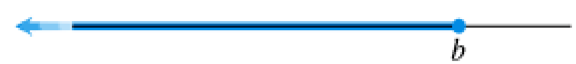
Intervals For $a, b \in \mathbb{R}$,

$$(a, b) = \{ x \in \mathbb{R} \mid a < x < b \}$$

whereas

$$[a, b] = \{ x \in \mathbb{R} \mid a \leqslant x \leqslant b \}.$$

TABLE 1.1 Types of intervals

	Notation	Set description	Type	Picture
Finite:	(a, b)	$\{x \mid a < x < b\}$	Open	
	$[a, b]$	$\{x \mid a \leq x \leq b\}$	Closed	
	$[a, b)$	$\{x \mid a \leq x < b\}$	Half-open	
	$(a, b]$	$\{x \mid a < x \leq b\}$	Half-open	
Infinite:	(a, ∞)	$\{x \mid x > a\}$	Open	
	$[a, \infty)$	$\{x \mid x \geq a\}$	Closed	
	$(-\infty, b)$	$\{x \mid x < b\}$	Open	
	$(-\infty, b]$	$\{x \mid x \leq b\}$	Closed	
	$(-\infty, \infty)$	$\mathbb{R}$ (set of all real numbers)	Both open and closed	

Intersections and Unions

Intersection of two intervals:

$(a, b) \cap (c, d) =$

$\{x \in \mathbb{R} \mid x \in (a, b) \text{ AND } x \in (c, d)\}$

$= \{x \in \mathbb{R} \mid a < x < b \text{ AND } c < x < d\}$

Union of two intervals:

$(a, b) \cup (c, d) =$

$\{x \in \mathbb{R} \mid x \in (a, b) \text{ OR } x \in (c, d)\}$

$= \{x \in \mathbb{R} \mid a < x < b \text{ OR } c < x < d\}$

Examples: unions & intersections

Simplify and give your answer in interval notation.

- $(1,3) \cup (2,4]$
- $[-4,\sqrt{2}) \cap [0,1)$
- $[\sqrt{3},5) \cap (1.8,7)$
- $[0,4) \cap \Big((-2,1) \cup [3,7)\Big)$

Recall, $\sqrt{2} \approx 1.41\ldots$ and $\sqrt{3} \approx 1.73\ldots$.

Examples: unions & intersection

Solutions:

- $(1,3) \cup (2,4] = (1,4]$
- $[-4, \sqrt{2}) \cap [0,1) = [0,1)$
- $[\sqrt{3}, 5) \cap (1.8, 7) = (1.8, 5)$
- $[0,4) \cap \Big((-2,1) \cup [3,7)\Big)$
  $= [0,1) \cup [3,4)$

PRE-CALCULUS
INEQUALITIES

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Understanding inequalities graphically

Consider the following functions:

$$f(x) = x^2 - 1$$
$$g(x) = (x - 1)^2$$

and the inequality $g(x) < f(x)$.

Also, notice the difference between solving for x in yesterday's tut question:

$$x^2 + 3x - 18 = 0$$

and

$$x^2 + 3x - 18 \geqslant 0$$

Rules for Inequalities

If a, b, and c are real numbers, then:

1. $a < b \Rightarrow a + c < b + c$
2. $a < b \Rightarrow a - c < b - c$
3. $a < b$ and $c > 0 \Rightarrow ac < bc$
4. $a < b$ and $c < 0 \Rightarrow bc < ac$
   Special case: $a < b \Rightarrow -b < -a$
5. $a > 0 \Rightarrow \dfrac{1}{a} > 0$
6. If a and b are both positive or both negative, then $a < b \Rightarrow \dfrac{1}{b} < \dfrac{1}{a}$

Solving inequalities

We will often make use of a number line or a table to determine the sign of the function on particular intervals. We use critical values (where a function is 0 or undefined) to determine the intervals.

Examples:

1. Solve for x: $1 + x < 7x + 5$
2. Solve for x: $x^2 < 2x$

Solving $x^2 < 2x$

$$x^2 < 2x$$

$$\therefore \quad x^2 - 2x < 0$$

$$\therefore \quad x(x-2) < 0$$

CVs: $x = 0, \ x = 2$

$+ \quad - \quad +$

$(x=-1) \quad 0 \quad (x=1) \quad 2 \quad (x=3)$

$\therefore \quad 0 < x < 2$ or $x \in (0,2)$

Solving inequalities continued

Find solutions to the following inequalities and write the solutions in interval notation.

1. $4 \leqslant 3x - 2 < 13$
2. $x^2 - 5x + 6 \leqslant 0$
3. $x^3 + 3x^2 > 4x$ (Don't divide by x!)
4. $\dfrac{x^2 - x - 6}{x^2 + x - 6} \leqslant 0$
5. $\dfrac{x^2 - 6}{x} \leqslant -1$ (Don't multiply by x!)

Solution to (5)

$$\frac{x^2-6}{x}+1\le 0 \quad \therefore \frac{x^2+x-6}{x}\le 0$$

$$\therefore \frac{(x+3)(x-2)}{x}\le 0 \qquad \text{CVs}: -3, 0, 2$$

	$x+3$	$x-2$	x	LHS
$(-\infty,-3)$	$-$	$-$	$-$	$-$
$(-3,0)$	$+$	$-$	$-$	$+$
$(0,2)$	$+$	$-$	$+$	$-$
$(2,\infty)$	$+$	$+$	$+$	$+$

$$x\in(-\infty,3]\cup(0,2]$$

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ABSOLUTE VALUE

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Absolute value

The **absolute value** of a number a, denoted by $|a|$ is the distance from a to 0 along the real line. A distance is always positive or equal to 0 so we have

$$|a| \geqslant 0 \text{ for all } a \in \mathbb{R}.$$

Examples

$$|3| = 3 \qquad |-3| = 3 \qquad |0| = 0$$

$$|2-\sqrt{3}| = 2-\sqrt{3} \qquad |3-\pi| = \pi - 3$$

In general we have

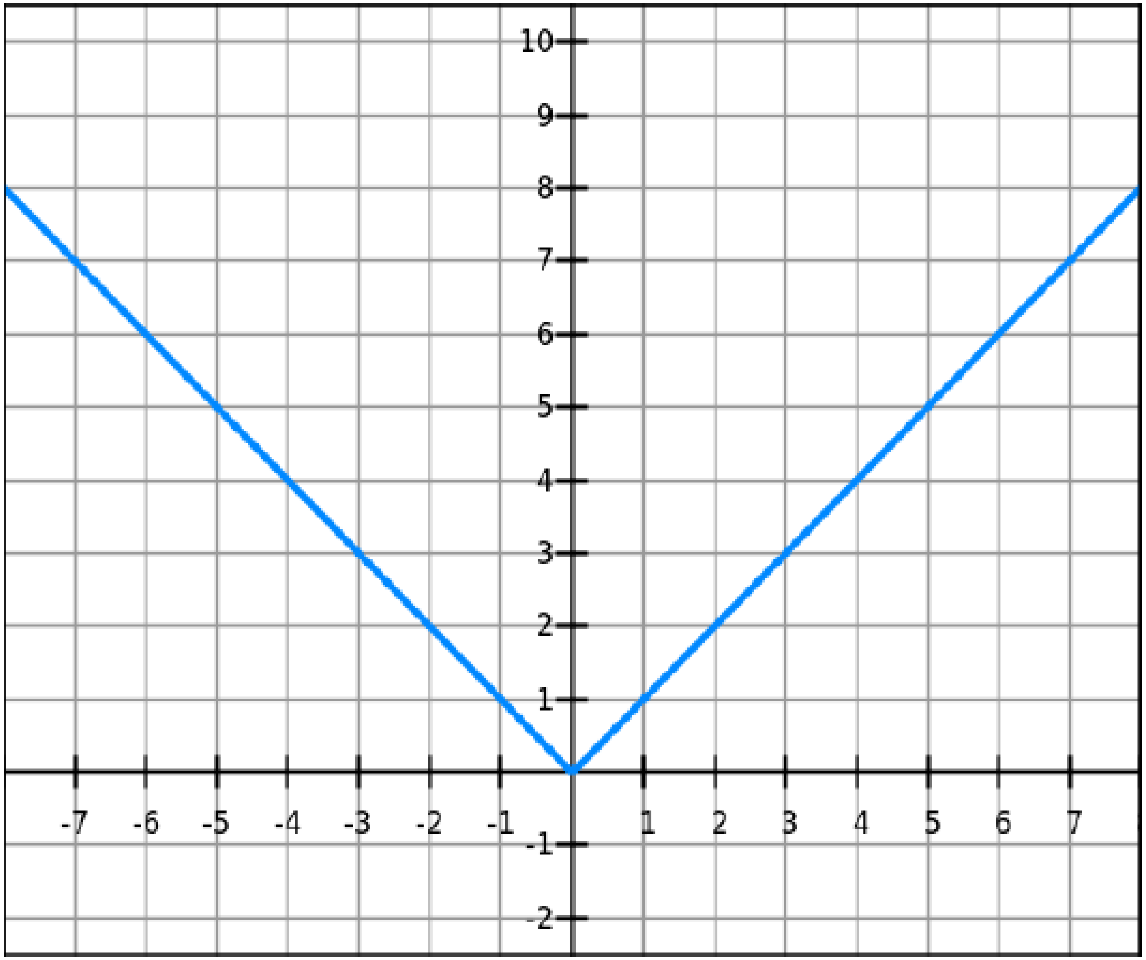
$$|a| = a \qquad \text{if } a \geqslant 0$$
$$|a| = -a \qquad \text{if } a < 0$$

We can write the absolute value function as a **piecewise defined function**.

$$|x| = \begin{cases} x & \text{if } x \geqslant 0 \\ -x & \text{if } x < 0 \end{cases}$$

We can also replace the x above with something more complicated.

Sketching $y = |x|$:



If $f(x) = |x|$, calculate $f(-5)$, $f(4)$ and $f(0)$.

- $f(-5) = -(-5) = 5$
- $f(4) = 4$
- $f(0) = 0$

Note: for any $a \in \mathbb{R}$, $|a| = \sqrt{a^2}$

Example

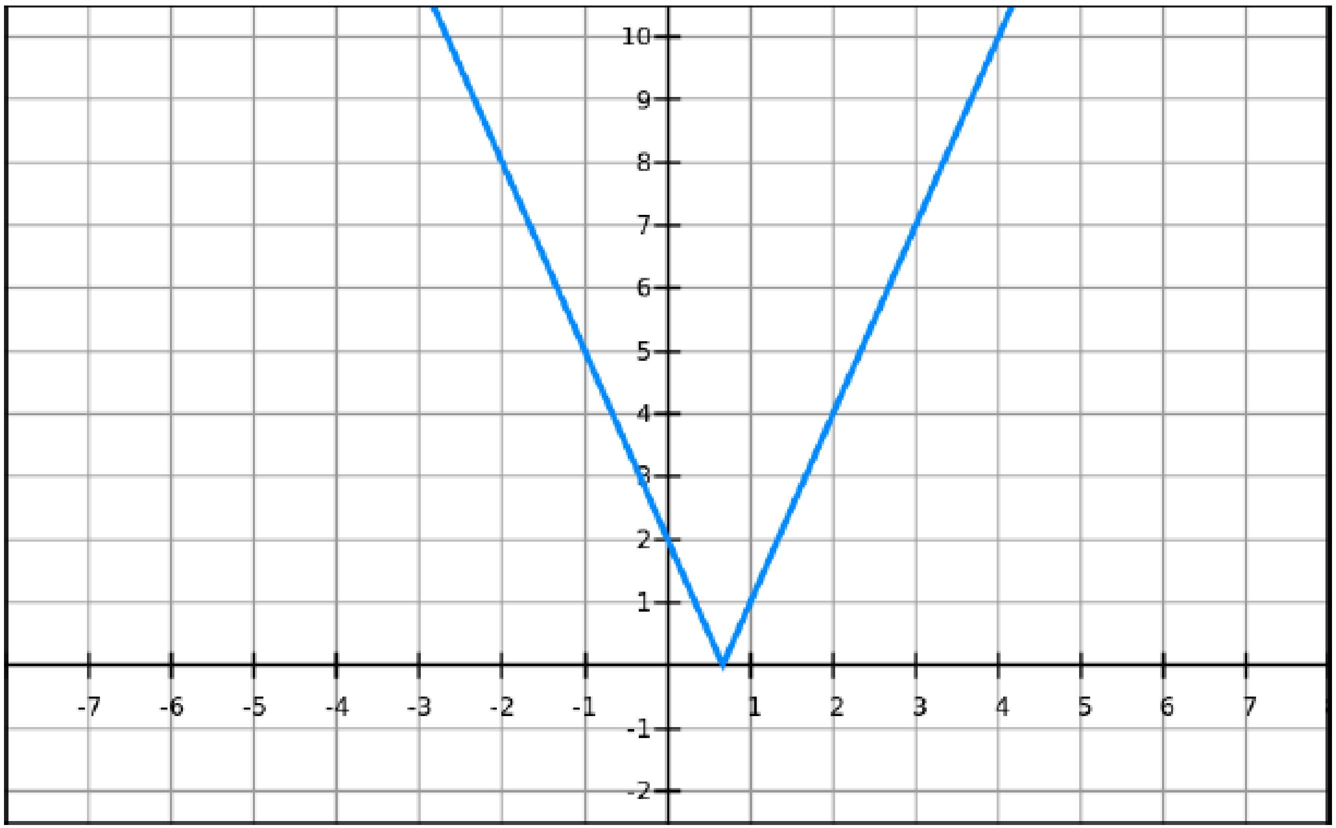
Write $|3x - 2|$ without using the absolute value symbol.

$$|3x - 2| = \begin{cases} 3x - 2 & \text{if } 3x - 2 \geqslant 0 \\ -(3x - 2) & \text{if } 3x - 2 < 0 \end{cases}$$

Hence

$$|3x - 2| = \begin{cases} 3x - 2 & \text{if } x \geqslant \frac{2}{3} \\ 2 - 3x & \text{if } x < \frac{2}{3} \end{cases}$$

Below is a sketch of $y = |3x - 2|$. Note how the function changes at $x = \frac{2}{3}$.



Properties of Absolute Values

Suppose $a, b \in \mathbb{R}$ and $n \in \mathbb{Z}$. Then

1. $|ab| = |a||b|$
2. $|\frac{a}{b}| = \frac{|a|}{|b|}$ $(b \neq 0)$
3. $|a^n| = |a|^n$

Let $a > 0$. Then

4. $|x| = a$ if and only if $x = a$ or $x = -a$.
5. $|x| < a$ if and only if $-a < x < a$.
6. $|x| > a$ if and only if $x > a$ or $x < -a$.

Example: Solve $|2x - 5| = 3$.

Solving absolute value inequalities:

- Solve: $|x-5|<2$.
- Solve: $|3x+2|\geqslant 4$.

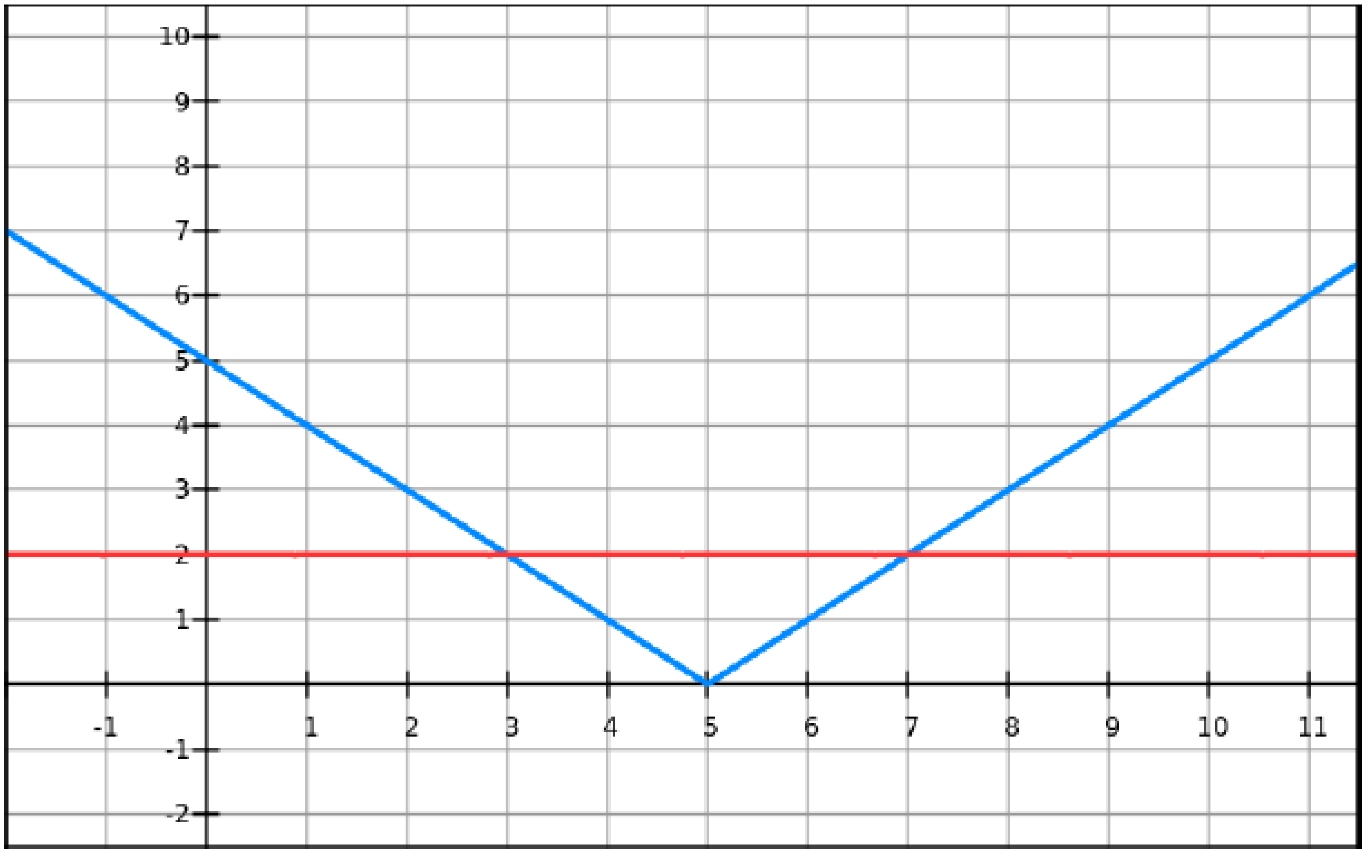
To solve the first inequality above we must solve

$$-2<x-5<2$$

For the second inequality we have

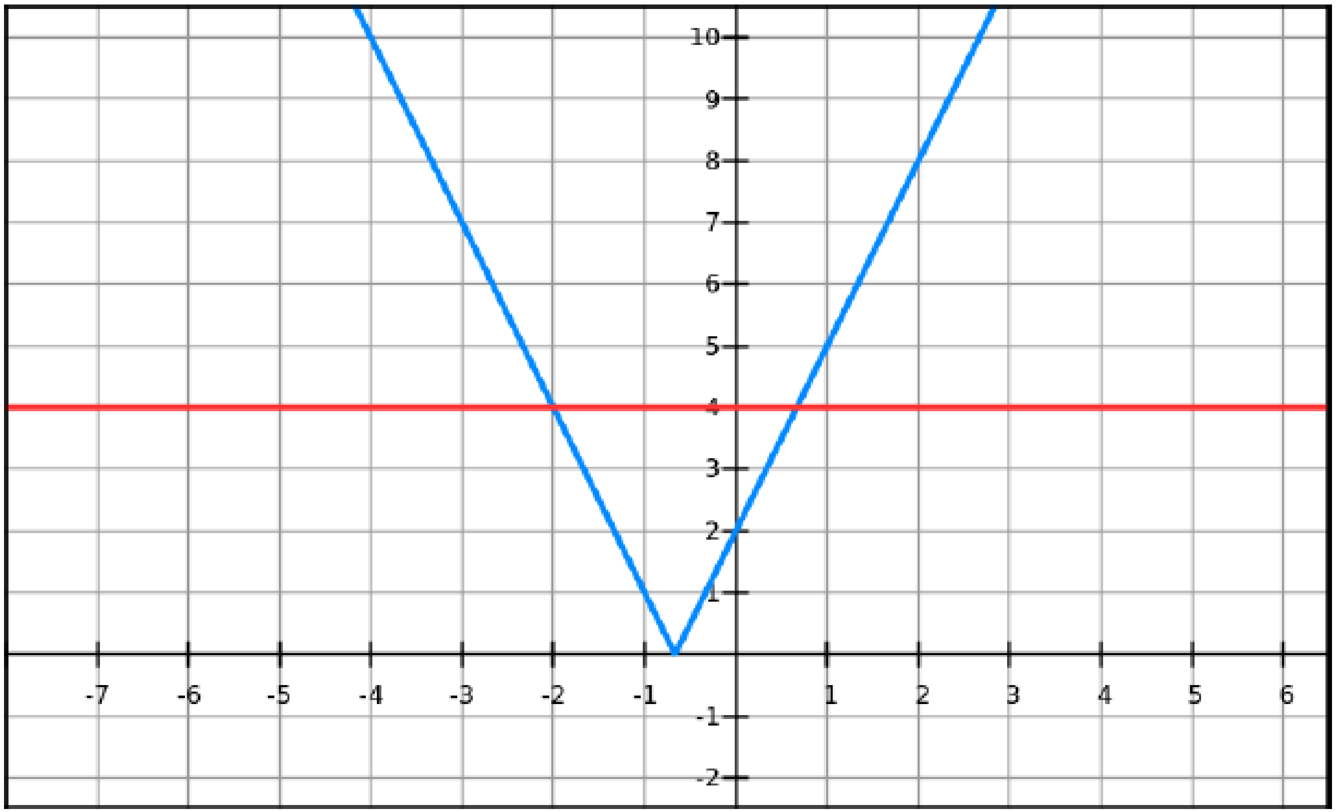
$$3x+2\geqslant 4 \qquad \text{OR} \qquad 3x+2\leqslant -4$$

Solving $|x - 5| < 2$ gives $x \in (3, 7)$.



Solving $|3x + 2| \geqslant 4$ gives

$$x \in (-\infty, -2] \cup [\tfrac{2}{3}, \infty)$$



Warm up

Let $a > 0$. Then

4. $|f(x)| = a$ if and only if $f(x) = a$ or $f(x) = -a$.
5. $|f(x)| < a$ if and only if $-a < f(x) < a$.
6. $|f(x)| > a$ if and only if $f(x) > a$ or $f(x) < -a$.

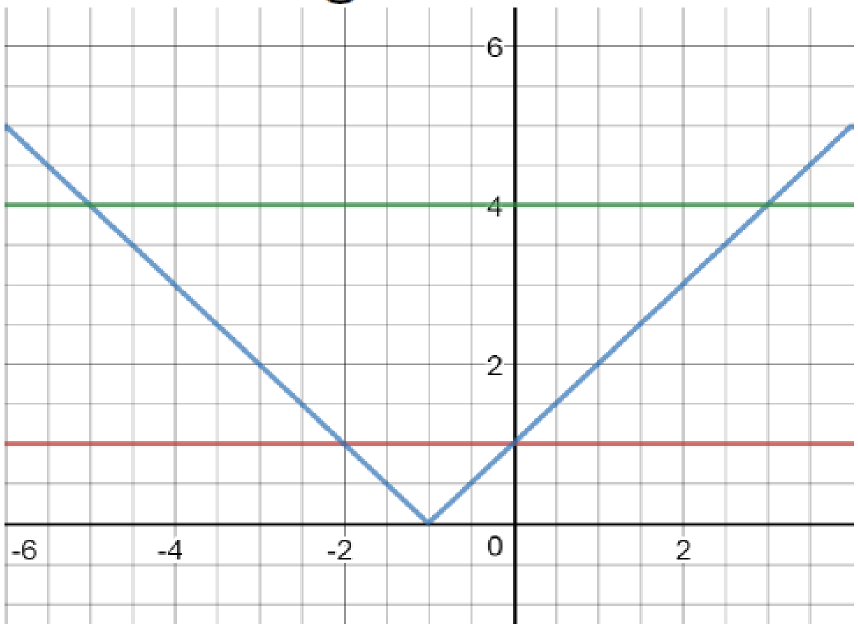
Now solve:

$$1 < |x + 1| < 4$$

The solution is the values of x that satisfy:

$$1 < |x+1| \quad \text{AND} \quad |x+1| < 4$$

Therefore $x \in (-5,-2) \cup (0,3)$. This algebraic solution agrees with the sketch:



The triangle inequality

$\boxed{\textit{If } a, b \in \mathbb{R}, \textit{ then } |a+b| \leqslant |a| + |b|.}$

How do we prove this?

First observe that $-|a| \leqslant a$ and $a \leqslant |a|$.

Similarly, $-|b| \leqslant b$ and $b \leqslant |b|$.

Applications:

Example: If $|x-4|<0.1$ and $|y-7|<0.2$, use the Triangle Inequality to estimate $|(x+y)-11|$.

Exercise: show that if $|x+3|<\frac{1}{2}$, then $|4x+13|<3$.

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INTRODUCTION TO FUNCTIONS

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Introduction to functions

Four examples of functions:

- The area of a circle depends on the radius: $A = \pi r^2$.
- Population of the world depends on time: $P(1950) = 2,560,000,000$.
- The cost of posting a package depends on the weight: $C(w)$.
- The vertical ground acceleration during an earthquake: $S(t)$.

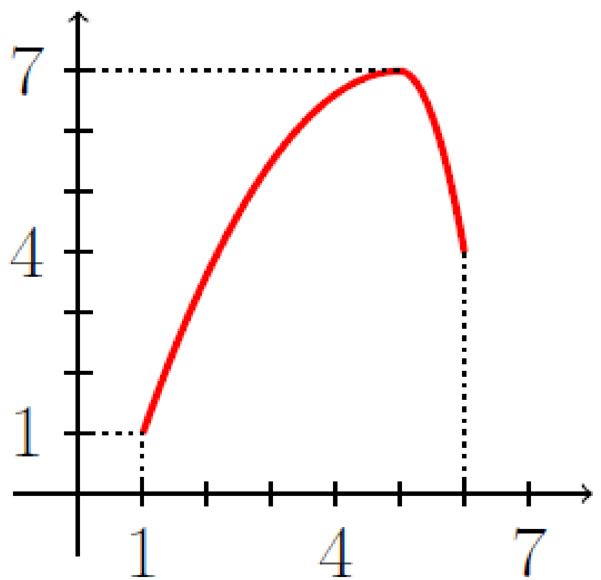
Domain and range of functions

A *function* f is a rule which assigns to each element x in a set D exactly one element, $f(x)$, in a set E.

The set D is the *domain* of f. The *range* of f is the set of all possible values of $f(x)$ as x varies through the domain.

A symbol representing an arbitrary element of the domain is called an *independent variable* and a symbol representing an arbitrary element of the range is a *dependent variable*. In the example of the circle: r is the *independent variable* while A is the *dependent variable*.

Graphs of functions: a common way of representing a function is by a graph. Formally, the graph of the function f is the set of ordered pairs $\{\, (x, f(x)) \mid x \in D \,\}$.



Dom(f)= [1,6] and Ran(f)=[1,7]

Examples

Sketch the following functions and find their domain and range:

(a) $f(x) = -3x + 4$

(b) $g(x) = x^2 - 2$

(c) $h(x) = \sec x$

Representations of functions

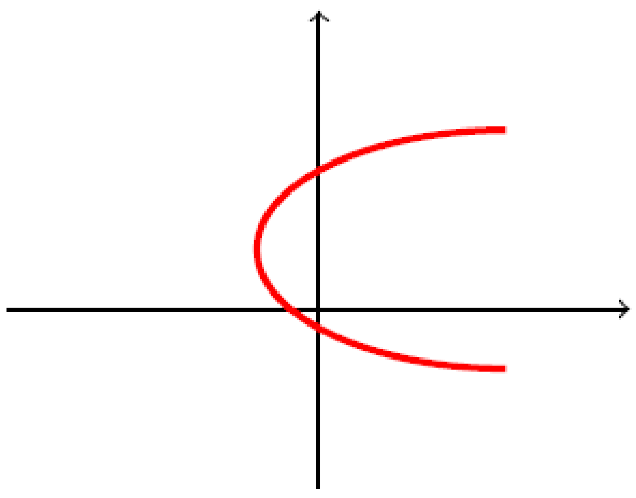
There are four ways to represent a function:

- Verbally (describe in words)
- Tables
- Graphically
- Algebraically

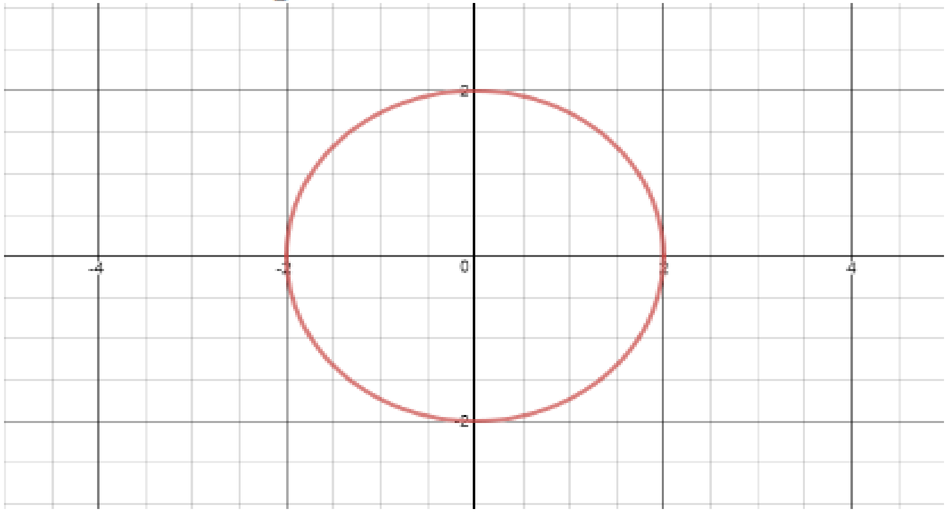
The vertical line test

How do we know if a curve is a function?

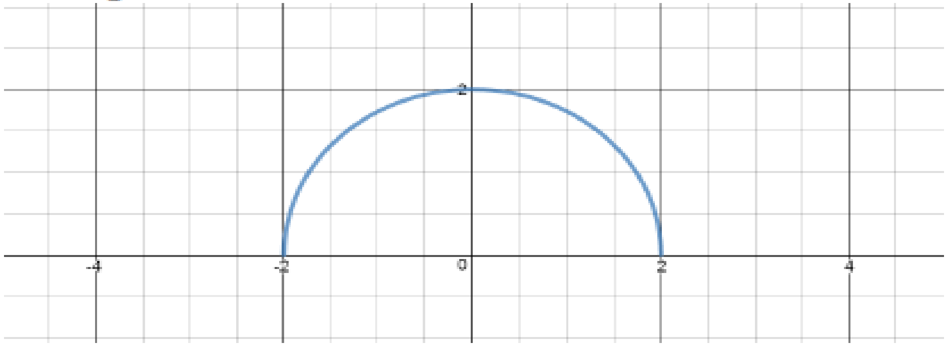
A curve in the xy-plane is a function of x if and only if *no vertical line intersects the curve more than once.*



The curve $x^2 + y^2 = 4$ is *not* a function:



The curve $y = \sqrt{4 - x^2}$ *is* a function:



Piecewise defined functions

We are already familiar with one example of a piecewise defined function, the absolute value function:

$$f(x) = |x| = \begin{cases} x & \text{if } x \geqslant 0 \\ -x & \text{if } x < 0 \end{cases}$$

Now sketch the function

$$f(x) = \begin{cases} -x+2 & \text{if } x \leqslant 1 \\ x^2 & \text{if } x > 1 \end{cases}$$

Example: step functions

Consider the cost of travelling on the Rea Vaya bus. Here x is the number of km travelled and $C(x)$ is in rands:

$$C(x) = \begin{cases} 7.5 & \text{if } 0 \leqslant x \leqslant 5 \\ 9 & \text{if } 5 < x \leqslant 10 \\ 11 & \text{if } 10 < x \leqslant 15 \\ 13 & \text{if } 15 < x \leqslant 25 \\ 14 & \text{if } 25 < x \leqslant 35 \\ 15 & \text{if } 35 < x \end{cases}$$

Symmetry in functions

If a function f satisfies

$$f(-x) = f(x) \qquad \text{for all } x \in D$$

then f is an *even* function.

Examples:

- $f(x) = x^2$
- $f(x) = \cos(x)$
- $f(x) = |x|$

Another way of defining an even function is to say that it is a reflection about the y-axis.

Symmetry in functions

A function is *odd* if

$$f(-x) = -f(x) \qquad \text{for all } x \in D$$

Examples:

- $f(x) = x$
- $f(x) = \sin(x)$
- $f(x) = x^3$

An odd function is a reflection about the origin. A necessary condition for a function f to be odd is that it must have $f(0) = 0$.

Increasing and decreasing functions

A function f is *increasing* on an interval I if whenever $x_1 < x_2$, we have $f(x_1) < f(x_2)$.

A function f is *decreasing* on an interval I if whenever $x_1 < x_2$ we have $f(x_1) > f(x_2)$.

Example: Is the function $f(x) = \cos x$ increasing, decreasing, or neither over the following intervals:

(a) $x \in [\pi, 3\pi/2]$
(b) $x \in [0, \pi/2]$
(c) $x \in [\pi/2, 3\pi/2]$

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TYPES OF FUNCTIONS

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Mathematical models

A *mathematical model* is a mathematical description, using a function or an equation, of a real-world problem.
If y is a linear function of x then the graph is a straight line:

$$y = mx + c$$

where m is the slope of the graph, and c is the y-intercept.

Example of a linear model: The length of a newly born snake is 10cm and after 3 months the length is 25cm. It grows the same amount each month.

(a) Express length as a function of time (in months) where the $D = [0, 12]$.

(b) Draw the graph of the length function.

(c) What is the length of the snake after 9 months?

<u>Functions from data:</u> If we don't have a function to work from, we can try to determine a function using empirical data.

Year	Number of registered cars in SA
2005	4,500,000
2006	4,670,000
2007	4,890,000
2008	5,100,000
2009	5,310,000
2010	5,540,000
2011	5,770,000
2012	5,900,000

For the last example we can try to find a function which fits the data by calculating the slope of the line which goes through the first and last points.

$$m = \frac{5,900,000 - 4,500,000}{2012 - 2005}$$

Therefore $m = 200,000$.

To be more accurate, we can use a statistical technique known as *linear regression*.

Polynomials

These are functions of the form:

$$f(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0$$

where $a_0, \ldots, a_n$ are constants. If the leading coefficient $a_n \neq 0$ then $f(x)$ is a polynomial of degree n.

- polynomial of degree $1 =$ linear function
- polynomial of degree $2 =$ quadratic function
- polynomial of degree $3 =$ cubic function

Power functions

These are functions of the form

$$f(x) = x^a$$

Note that a can be

- a positive integer
- $a = \frac{1}{n}$ where n is a positive integer
- a negative integer, so $f(x) = \frac{1}{x^a}$

Rational functions

A rational function f is a ratio of two polynomials

$$f(x) = \frac{P(x)}{Q(x)}$$

Example:

$$f(x) = \frac{2x^4 - x^2 + 1}{x^2 - 4}$$

Algebraic functions

An algebraic function is one that can be formed by using the algebraic operations of addition, subtraction, multiplication, powers, division and taking roots.
Note: any rational function is automatically an algebraic function.

Examples:

$$f(x) = \sqrt{x^2 + 1} \qquad h(x) = \frac{x^4 - 16x^2}{x + \sqrt{x}}$$

Trigonometric functions

Functions that express the ratio between x, y and r when angles are plotted on the xy-plane.

- $\sin x$
- $\cos x$
- $\tan x$
- $\csc x$
- $\sec x$
- $\cot x$

Exponential functions

These are functions of the form

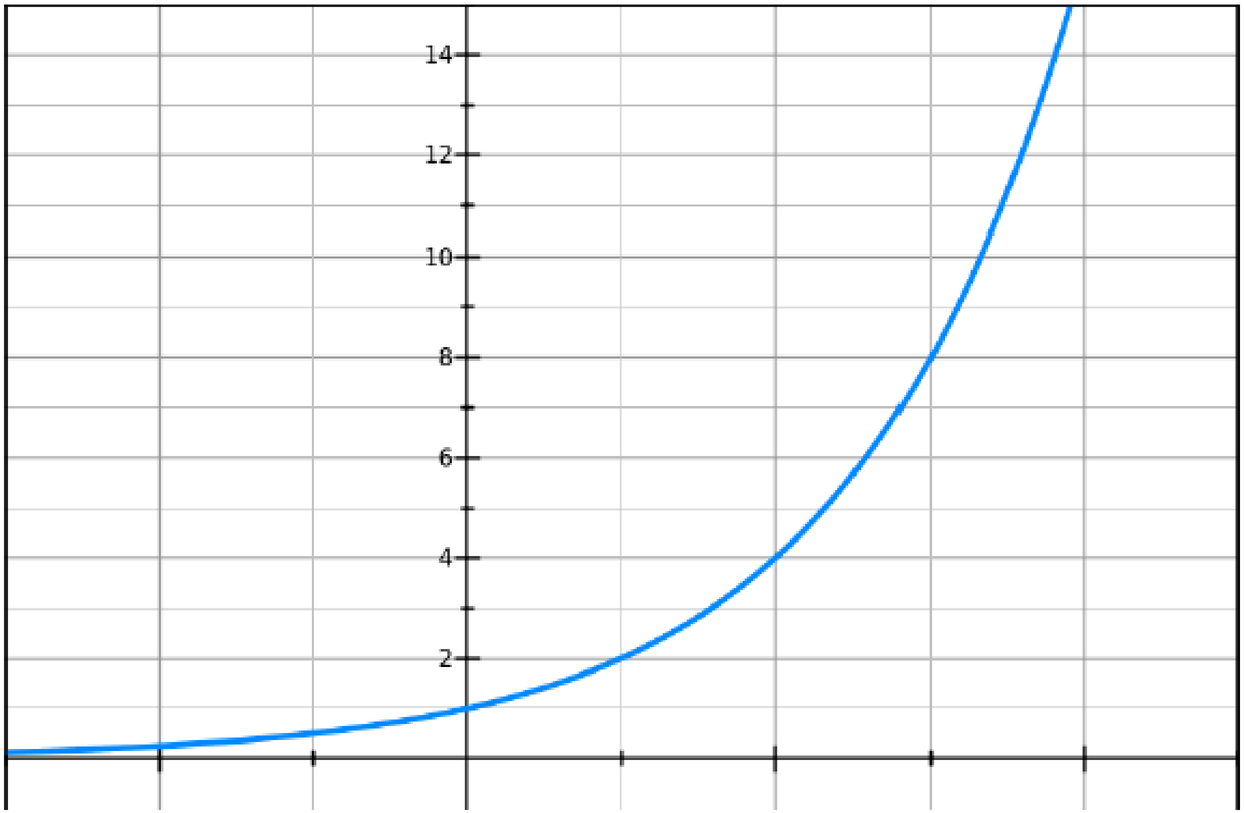
$$f(x) = a^x$$

where $a \geqslant 0$ is constant.

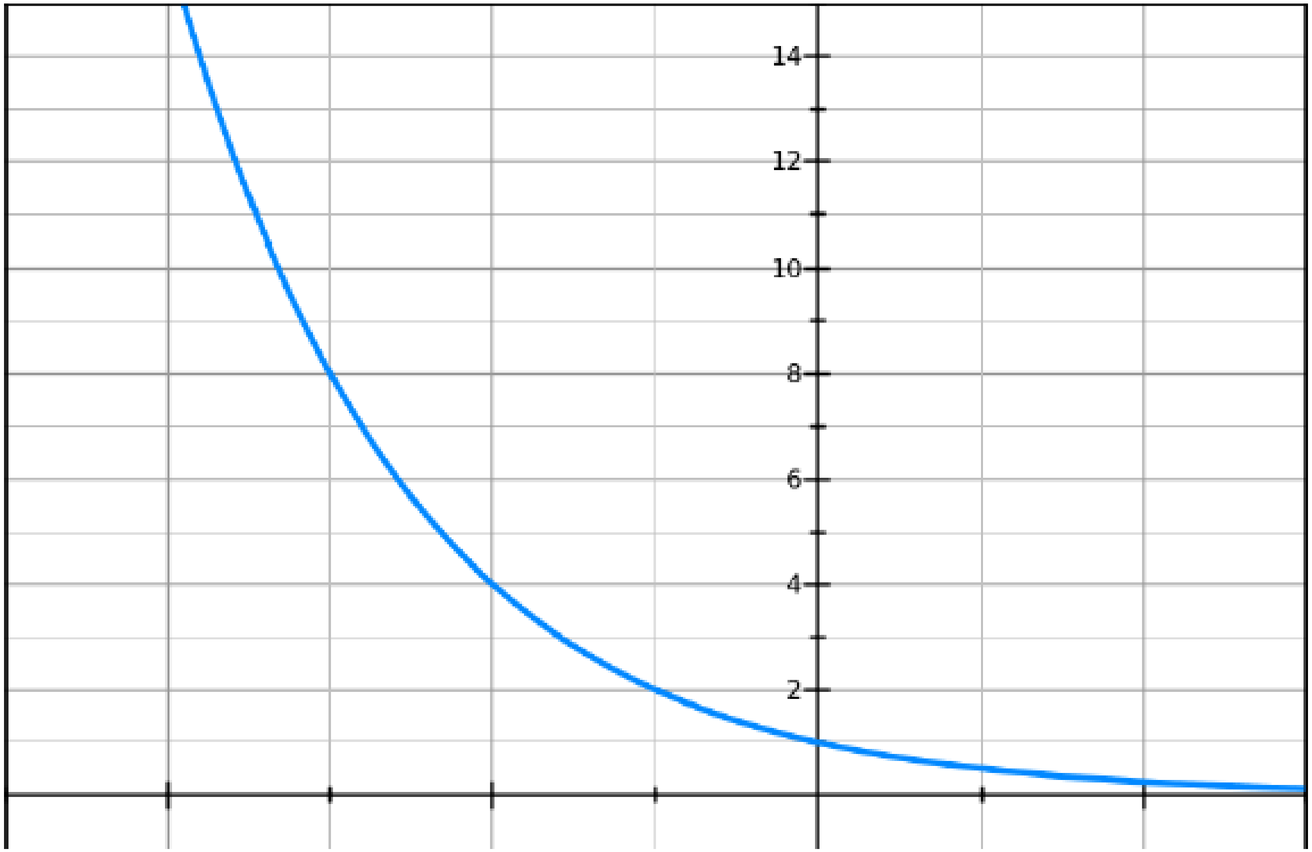
Examples:

- $f(x) = 2^x$
- $f(x) = 0.5^x$
- $f(x) = e^x$

$$f(x) = 2^x$$



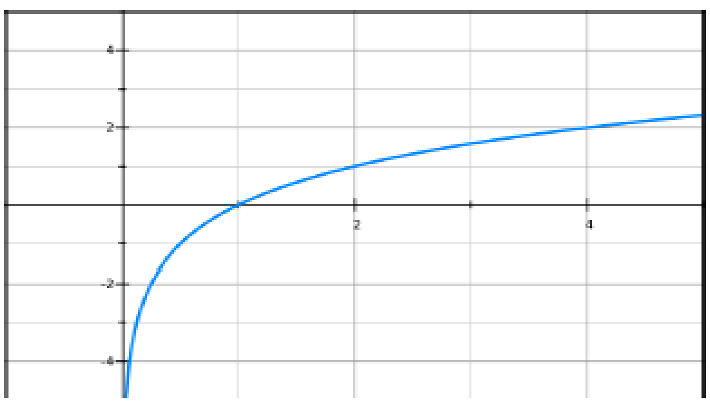
$$f(x) = 0.5^x$$



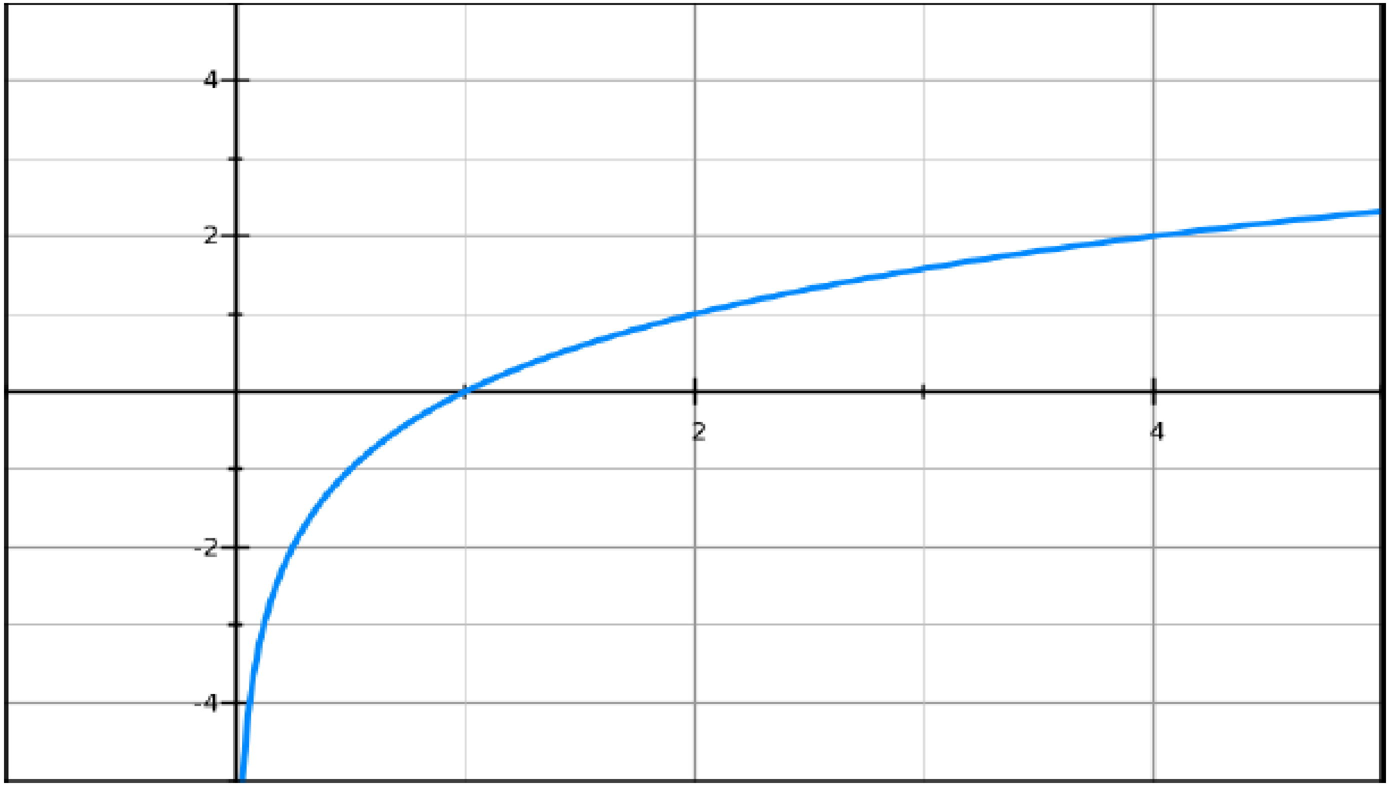
Logarithmic functions

$f(x) = \log_a x$ where $a \geqslant 0$ is a constant (also known as the *base*). Logarithmic functions are the inverse of exponential functions. That is,

$$\text{if } y = a^x \quad \text{then} \quad \log_a y = x.$$



$$f(x) = \log_2 x$$



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